

Field Case Study: North Sea HPHT Lost Circulation and Induced Kick (2015)

1. Drilling Profile & Geology

While drilling the 8-1/2 inch section of an offshore High-Pressure High-Temperature (HPHT) exploratory well at 16,800 feet true vertical depth (TVD), total lost circulation was encountered in a fractured chalk reservoir.

2. Lost Circulation Event

A sudden drop in pit volume of 180 barrels occurred within 12 minutes. The equivalent circulating density (ECD) of 15.2 ppg exceeded the formation breakdown pressure, creating open fractures and rapidly lowering the liquid hydrostatic head in the annulus.

3. Secondary Influx (Kick)

As the fluid level dropped by over 1,200 feet, bottom-hole pressure fell below reservoir pore pressure (14.6 ppg equivalent), resulting in an immediate 35-barrel gas kick. Surface telemetry showed a rapid increase in standpipe pressure and gas units on the mud loggers.

4. Well Control Response

The crew executed a soft shut-in: spaced out drill string, stopped rotary pumps, and closed the upper annular preventer. A high-density crosslinked polymer Lost Circulation Material (LCM) pill was pumped down drill pipe, followed by the Driller's Method circulation to remove gas influx.